

Asymmetric Excitation

The CPMG-conditions require, that the phase of the excitation pulse is orthogonal to that of the refocusing pulses. The most simple way to meet this condition is to use phase of $+90^\circ$ for the excitation pulse and 0° for the refocusing pulses. It should be noted that due to the frequency invariance of the Bloch equations this can be generalized to

$$\pm 90^\circ \pm \alpha - \pm 2\alpha - \pm 4\alpha - \dots - \pm 2ne \alpha,$$

where α represents an arbitrary phase offset and ne indicates the echo number. For $\alpha = 90^\circ$ this leads to the commonly used alternating phase scheme

$$0^\circ - 180^\circ - 0^\circ - 180^\circ - 0^\circ - 180^\circ - 0^\circ - \dots$$

CP-conditions leading to destructive interference of the different refocusing pathways occur, if the excitation pulse is in phase with the refocusing pulse. This is illustrated in Fig.D1 for simulations using sinc-pulses for both the excitation and the refocusing pulses.

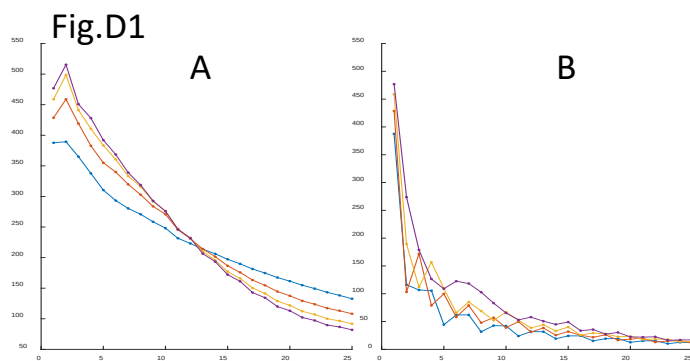


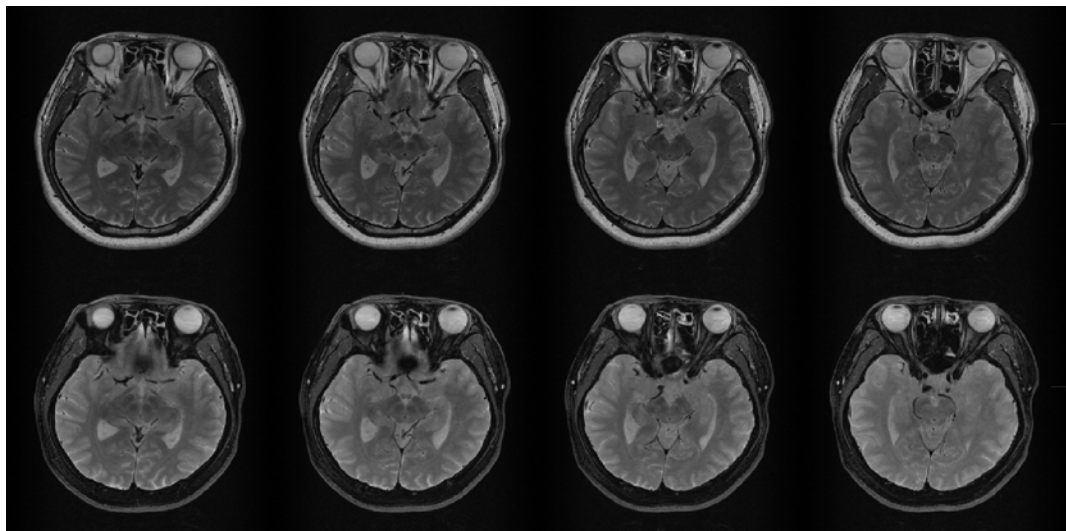
Fig.D1 Signal decay for gray matter under CPMG(A) and CP(B) conditions for gray matter and refocusing flip angles 60(blue), 90(red), 120 (yellow), and 150(magenta) degrees.

simulations using sinc-pulses for both the excitation and the refocusing pulses.

This can be used for chemical shift selective signal attenuation eg. in fat suppression. If the time between the excitation pulse and the first refocusing pulse is changed by a time interval dTE , where dTE leads to a phase evolution of 90° for fat signals, then fat will effectively 'see' a CP-sequence with low signal intensity,

whereas on-resonance spins show the beneficial CPMG-signal decay. dTE should be $1/(4\Delta f)$ to achieve this condition, which corresponds to $\sim 0.6-0.7$ ms at 3T. For $dTE=1/(2\Delta f)$, fat will be out-of-phase to water. Fat-selective imaging is achieved using $dTE=1/(4\Delta f)$ combined with CP-conditions for the pulse phases, fat will then evolve to 'see' CPMG-conditions. Note that the signal in CP-conditions is high at the first echo, so this will not work at short TE_{eff} .

Fig.D2



As with other chemical shift-dependent fat suppression schemes, signal in areas of strong local field inhomogeneity is also affected. Fig.D2 shows results of measurements with echo spacing 10 ms, TE_{eff} 80 ms and $dTE=0.7$ ms. Note the good suppression of orbital fat. Signal above subarachnoidal spaces is also attenuated due to local susceptibility effects..

The intrinsic fat suppression offered by the asymmetric excitation scheme is especially suitable for applications at high fields, where B1-inhomogeneity may not offer good fat suppression by shift selective pulses, it also saves RF-power by omitting the necessity of additional fat suppression pulses.